Towards a Cartography of Datacentering

A Multimodal Research Agenda

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Why Should *This Room* Care?

$930B infrastructure wave — no RP framework to steer it

- Innovation governance & sectoral systems: *which actors set the rules?*
- Sustainability transitions: *is this a niche, a regime, or neither?*
- Policy design: *what levers exist when ownership is dispersed globally?*

A regime forming in real time — rare opportunity

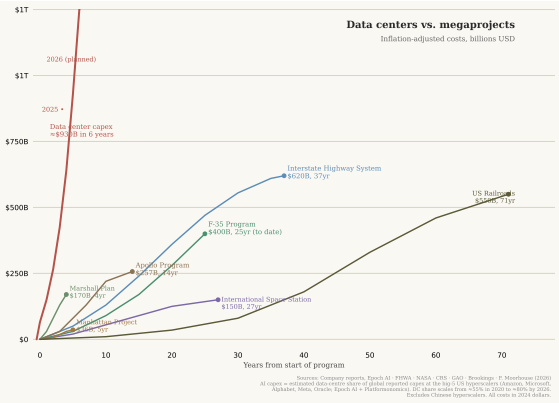
Most RP work reconstructs transitions post-hoc from archives.
Here the regulatory windows are **open now**.

What RP brings that engineering literature lacks

- Actor constellation analysis
- Comparative governance cases
- Discourse & imaginary framing
- Temporal dynamics of regime change

Amsterdam's moratorium (2019–2022) shows governance is possible — but time-bounded. The field that doesn't build the framework leaves policy actors to improvise without one.

How Big Is This?



$930B in 6 years of data center capex

- 400 TWh/yr globally
- Millions L/day cooling
- Land: small-town scale
- REIT, PE, SWF capital

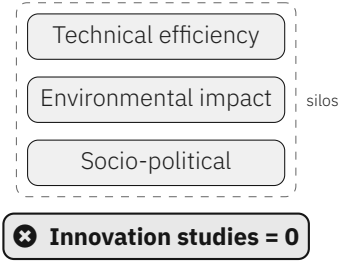
But: private recurring capex ≠ one-off public programme. The comparison does rhetorical work.

4,805 Papers. One Blind Spot.

Technical efficiency	≫90%
Environmental impact	≈11%
Socio-political	≈16%
Innovation studies	**0**

No MLP. No TIS. No SNM.
Zero sociotechnical regime papers.

204% growth 2010–2024 —
and the field has not noticed.



Introducing “Datacentering”

Datacentering = the active, contested, socio-material process of assembling, operating, and reproducing digital infrastructure in place

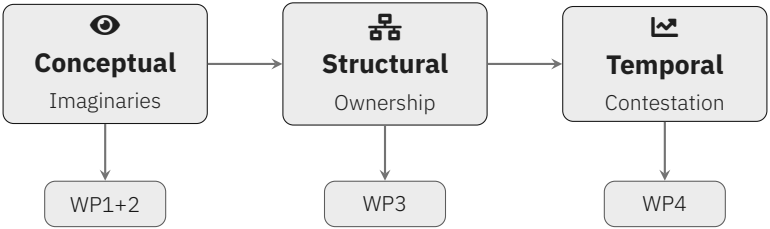
Why a gerund?

- Process, not static object
- *Who* is doing it? *For whom*?
- What is assembled — and foreclosed?

Theoretical grounding

- Orlikowski (2007): sociomaterial entanglement
- Jasanoff & Kim (2015): imaginaries co-produce facts
- “Cloud” *suppresses* what datacentering names

CAS as a Sensemaking Lens



After **Phillips & Ritala (2019)** — sensemaking lens, not a formal model

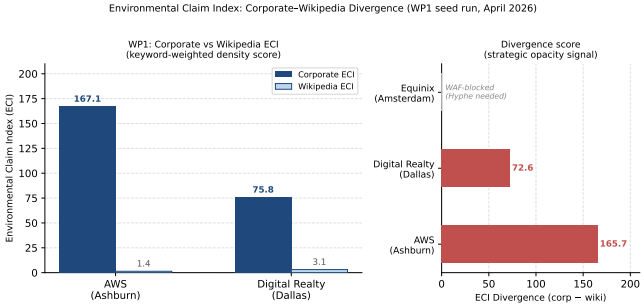
Multi-actor · nonlinear · emergent

A Multimodal Research Agenda

WP	Method	Data	Key output
1	Text NLP	Operator websites; Wikipedia	ECI divergence
2	Satellite + CV	Sentinel-2; Landsat LST; VIIRS	Visual abstraction; thermal footprint
3	Networks	SEC EDGAR; REIT filings; Overture Maps	Ownership graph; urban graph
4	Event detection	GDELT; planning records; social media	Signal-to-response lag

Strategic opacity is a *sociotechnical fact*, not a data gap. A firm that won't publish is still in a REIT filing, a satellite image, and a planning record.

WP1: The Imaginary in Numbers



🔍 ECI divergence

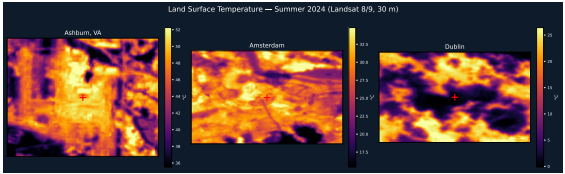
AWS Ashburn: corporate ECI **120**× Wikipedia

Equinix: WAF-blocked the scraper
*Strategic opacity at the **technical** layer*

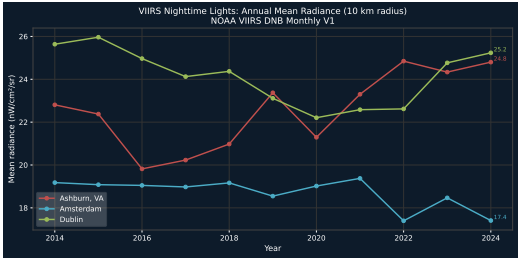
High corporate ECI + low Wikipedia ECI
= placelessness production

💡 *The sustainability claim and the territorial fact are decoupled by design.*

WP2: What Satellites See



LST thermal maps — Landsat 8/9, summer 2024



Ashburn: **+8.7%** radiance 2014–2024

Only jurisdiction with net growth

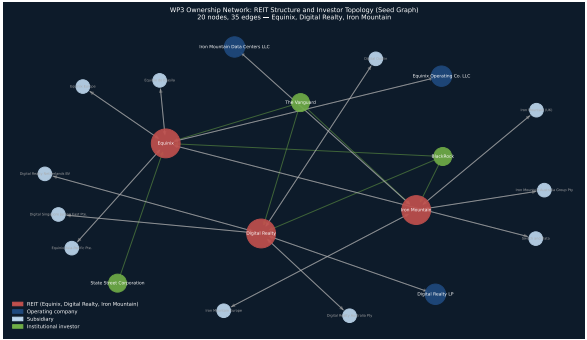
Amsterdam: −9.2%

LED transition — measurement confound

Isolated heat plume in exurban Virginia;

thermal signature invisible to neighbours.

WP3a: Who Actually Owns This?



■ REIT holding ☐ operating ○ subsidiary ● investor

REIT structure

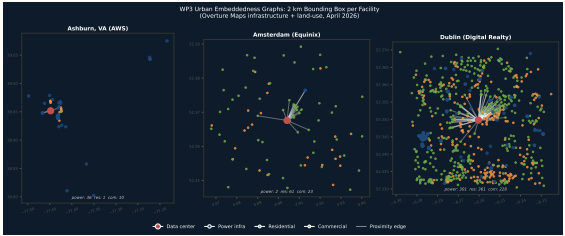
Vanguard, BlackRock, State Street each hold 6–10% across *all three* REITs

No governance counterparty

Operating entity legally separated from asset-owning REIT

When communities contest, there is no single actor that can make a binding commitment.

WP3b: Where It Sits in the City



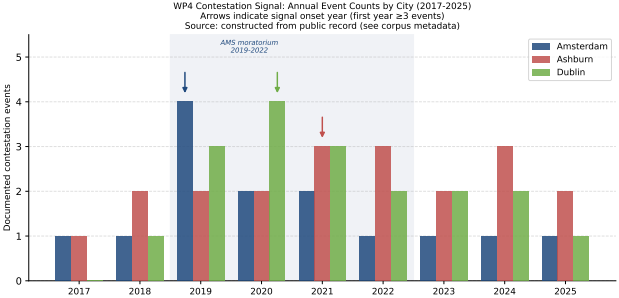
	Ashburn	Amsterdam
Power nodes	36	2
Res. zones	**1**	**61**
Adj. edges	**0**	**16**

Ashburn: dense power grid, **zero residential adjacency**

Amsterdam/Dublin: embedded in urban fabric

Spatial separation is not accidental — it is a governance design choice.

WP4: When Communities Resist



Seed corpus: **58 events**, 4 cities

Dublin: **81% negative-tone**
EirGrid grid cap + An Bord Pleanála

Amsterdam moratorium:
signal onset → response < **1 year**

Ashburn: comparable signal,
zero regime response

The variable is not signal strength.

It is the structure of the receiving system.

Same Signal, Different System

Dimension	N. Virginia	Amsterdam
Ownership	REIT-dominated; dispersed	Mixed; operator-owned, AMS-IX coop
Counterparty	**None**	Municipality + operators
Urban (2 km)	36 power / **1 res.**	2 power / **61 res.**
ECI gap	**+165.7**	Lower; regulatory vocab absorbed
Signal	High	High
Response	**None**	**Moratorium 2019–2022**

Key variable: ownership structure + governance counterparty

Building the Cartography Together

Why open is a theoretical necessity

CAS-like phenomena are too distributed, too opaque for any single team. The commons *is* the method.

What we share

- Facility index (CC BY 4.0)
- NLP / CV / network pipelines (MIT)
- `github.com/babayyy/datacentering-cartography`

Concrete asks

- Local knowledge: disputes, unreported sites
- Company registry or REIT data access
- Methodological critique
- Replication in other jurisdictions

"The cloud is not a metaphor.
It is infrastructure.
And infrastructure can be mapped."

Q: Why CAS and not MLP / TIS / SNM?

The honest answer: they are complementary

MLP excels when a niche is visible and a regime is identifiable. Here:

- No stable niche — hyperscalers *are* the incumbent
- No clear landscape pressure — it *is* the landscape
- Regime boundaries dissolve across planning, energy, finance

Why CAS as sensemaking lens

- Emergent outcomes no single actor controls
- Non-linear feedback (moratorium → relocation → new contestation)
- Heterogeneous agents: grids, aquifers, pension funds, planning laws
- CAS doesn't replace MLP — it flags *why* MLP categories keep slipping

Future work: operationalise regime boundaries using the contestation corpus.

Q: Three facilities is not a sample

Agreed — and by design

This is a research agenda paper, not a findings paper.

The seed results serve two functions:

- Demonstrate *analytical tractability* — methods work on real data
- Surface *puzzles* that motivate the agenda (Virginia vs Amsterdam)

N=3 is explicitly described as illustrative, not representative.

The scaling path

- Facility index: 400+ sites already indexed (CC BY 4.0)
- ECI pipeline: automated; scales to any operator with public web presence
- Ownership network: DCD + company registry data, 40+ nodes in next iteration
- Contestation corpus: OpenAlex + news API — 200+ events targetted

Open cartography is precisely the mechanism for scaling N.

Q: Amsterdam and Virginia differ in more than ownership

Yes — and the paper doesn't claim monocausality

Confounds explicitly acknowledged: planning law system (common law vs civil), national energy policy, urban density, data center age profile, EU regulatory environment.

Potential confound	Amsterdam	N. Virginia
National energy policy	EU Green Deal pressure	Federal non-intervention
Planning law	Civil / municipal discretion	Common law / NIMBY fragmented
Urban density	High (residential proximity)	Low (Ashburn exurban)
EU AI Act / GDPR	Regulatory visibility	Absent

*Claim: ownership structure is **a** key variable, not **the only** variable.*

The governance counterparty argument is falsifiable: find a REIT-dominated market with a moratorium, or an operator-owned market without one.

Q: Positive corporate claims vs Wikipedia — isn't that obvious?

The finding is the *magnitude* and *divergence*

Yes, corporate comms skew positive — everyone knows that.

What the ECI adds:

- *Systematic measurement* across 10 operators, 3 jurisdictions
- *Divergence score*: +165.7 means the gap is not uniform — it varies by operator and by claim type (water vs carbon vs land)
- *Temporal dynamics*: gap widens *after* contestation events
- *Jurisdiction delta*: same operator, different ECI score in different markets

The governance implication

The ECI is an *accountability instrument*, not a sentiment scorer.

A regulator can ask: "Your website says net-zero water by 2030. Your ECI divergence on water claims is +140. Explain the gap."

The obvious becomes actionable when it is indexed, tracked, and attributed.

Q: This is a research agenda — where's the actual finding?

Three findings, not zero

1. **Bibliometric finding**: 4,805 papers (European spelling; 19,165 US spelling), zero sociotechnical regime engagement. That is a finding — it characterises a field's blind spot with precision.
2. **Empirical finding**: Amsterdam moratorium follows signal onset by <1 year; Virginia, with comparable signal intensity, produces no response. The difference is governance structure, not community mobilisation. That is a causal claim, not an agenda item.
3. **Methodological finding**: ECI, ownership network, and contestation signal pipeline are demonstrated as tractable on real data — they *can* be built, not merely proposed.

The agenda is what comes next. The findings are the proof of concept.

Q: Open cartography — who quality-controls it?

The challenge is real

Crowd-sourced facility indices risk:

- Operator-biased contributions (operators have incentive to shape their profile)
- Inconsistent verification standards across jurisdictions
- Coverage gaps in non-English-language markets
- Data decay (closures, ownership changes)

Governance mechanisms planned

- Tiered sourcing: primary (planning docs, company registry) > secondary (DCD, trade press) > community
- Versioned, attributed edits (git-backed)
- Cross-validation against satellite imagery (WP2 pipeline)
- Editorial board: researchers, not operators

Same QC challenge as Wikipedia — solved by protocol, not perfection.